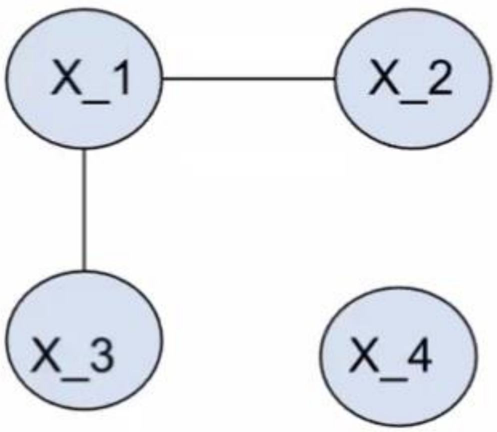
Q1)

[Review]:
- Covariance matrix $\Rightarrow$ marginal independence.
- Inverse covariance matrix $\Rightarrow$ conditional independence.

Graphical model is an easy way to help answer the questions.



(a) **[5 points]** Are X_3 and X_4 correlated?
<u>**Hint:**</u> Look at $\Sigma_{3,4} = 0$.

(b) **[5 points]** Are X_3 and X_4 conditionally correlated given the other variables? That is, does $cov(X_3, X_4|X_1, X_2)$ equal to zero?
<u>**Hint:**</u> Look at $Q_{34} = 0$ and so $cov(X_3, X_4|X_1, X_2) = 0$.

(d) **[5 points]** Assume that $Y = [Y_1, Y_2]^T$ is defined by:

$$Y_1 = X_1 + X_4$$

$$Y_2 = X_2 - X_4$$

Please calculate the covariance matrix of Y.

<u>**Hint:**</u> Rewrite Y as a linear tranformation of X:

$$Y = [Y_1, Y_2]^T = AX + B$$

$$Y_1 = X_1 + X_4 = 1 * X_1 + 0 * X_2 + 0 * X_3 + 1 * X_4$$
$$Y_2 = X_2 - X_4 = 0 * X_1 + 1 * X_2 + 0 * X_3 - 1 * X_4$$

Therefore, the covariance matrix of Y is:

$$Cov(Y) = Cov(AX) = A\Sigma A^T$$

Q2)

(a) **[5 points]** Assume we run EM starting from an initialization of $\mu_1 = -2$ and $\mu_2 = 2$. Please decide the value of μ_1 and μ_2 at the next iteration of EM algorithm. (You may find it handy to know that $1/(1 + exp(-4)) \approx 0.98$).?

Hint:

- First, run the E-step to calculate the probability of each data point belonging to each cluster.

$$\begin{aligned} Pr(x^{(1)}|\mu_1) &= \frac{1/2\phi(x^{(1)}|-2,1)}{1/2\phi(x^{(1)}|-2,1) + 1/2\phi(x^{(1)}|2,1)} \\ &= \frac{exp(-(-1+2)^2/2)}{exp(-(-1+2)^2/2) + exp(-(-1-2)^2/2)} \\ &\approx 0.98 \end{aligned}$$

and:

$$\begin{aligned} Pr(x^{(1)}|\mu_2) &= \frac{1/2\phi(x^{(1)}|2,1)}{1/2\phi(x^{(1)}|-2,1) + 1/2\phi(x^{(1)}|2,1)} \\ &= \frac{exp(-(-1-2)^2/2)}{exp(-(-1+2)^2/2) + exp(-(-1-2)^2/2)} \\ &\approx 0.02 \end{aligned}$$

Similarly, calculate $Pr(x^{(2)}|\mu_1)$ and $Pr(x^{(2)}|\mu_2)$.

- Then, run the M-step to update the parameters μ_1 and μ_2:

$$\mu_{1_{new}} = Pr(x^{(1)}|\mu_1) * x^{(1)} + Pr(x^{(2)}|\mu_1) * x^{(2)} \approx -0.96$$

$$\mu_{2_{new}} = Pr(x^{(1)}|\mu_2) * x^{(1)} + Pr(x^{(2)}|\mu_2) * x^{(2)} \approx 0.96$$

(b) **[5 points]** Do you think EM (when initialized with $\mu_1 = -2$ and $\mu_2 = 2$) will eventually converge to $\mu_1 = -1$ and $\mu_2 = 1$ (i.e., coinciding with the two data points). Please justify your answer using either your theoretical understanding or the result of an empirical simulation.

Hint:
Should be a no since if you keep running the EM algorithm, they're gonna converge to 0. For this question, you can either code it up and run the simulation or just provide a theoretical explanation by proving that $\mu_1 = -1$ and $\mu_2 = 1$ cannot be a fixed point.

(c) **[5 points]** Please decide the fixed point of EM when we initialize it from $mu_1 = \mu_2 = 2$.

Hint:
Use part a) and b) as a starting point and then just prove that $\mu_1 = \mu_2 = 0$ is a fixed point. by running one more iteration of the EM algorithm and showing that it doesn't update the means.

(d) **[5 points]** Please decide the fixed point of K-means when we initialize it from $\mu_1 = -2$ and $\mu_2 = 2$.

Hint:
- First iteration: $x^{(1)}$ is closer to μ_1 and $x^{(2)}$ is closer to μ_2.
- So each data point is assigned to a different cluster.
- $\mu_1 = x_1 = -1$ and $\mu_2 = x_2 = 1$.
- Then, no update since each cluster has only one data point for next iteration.